Identification of an Unknown: “B”

Savannah Creel

30 November 2023

CHEM 2081 - 007

Introduction

In this lab report, an unknown compound is identified using different data sets including Nuclear Magnetic Resonance (NMR), Mass Spectroscopy (MS), and Infrared Spectroscopy (IR). These instruments are very important because they are used to identify compounds. IR Spectroscopy is used to determine which functional groups are in a molecule, while NMR uses protons to identify a compound. It is important for chemists to know how to interpret the graphs produced by these instruments because it helps identify unknown compounds and can also be used to monitor chemical reactions.

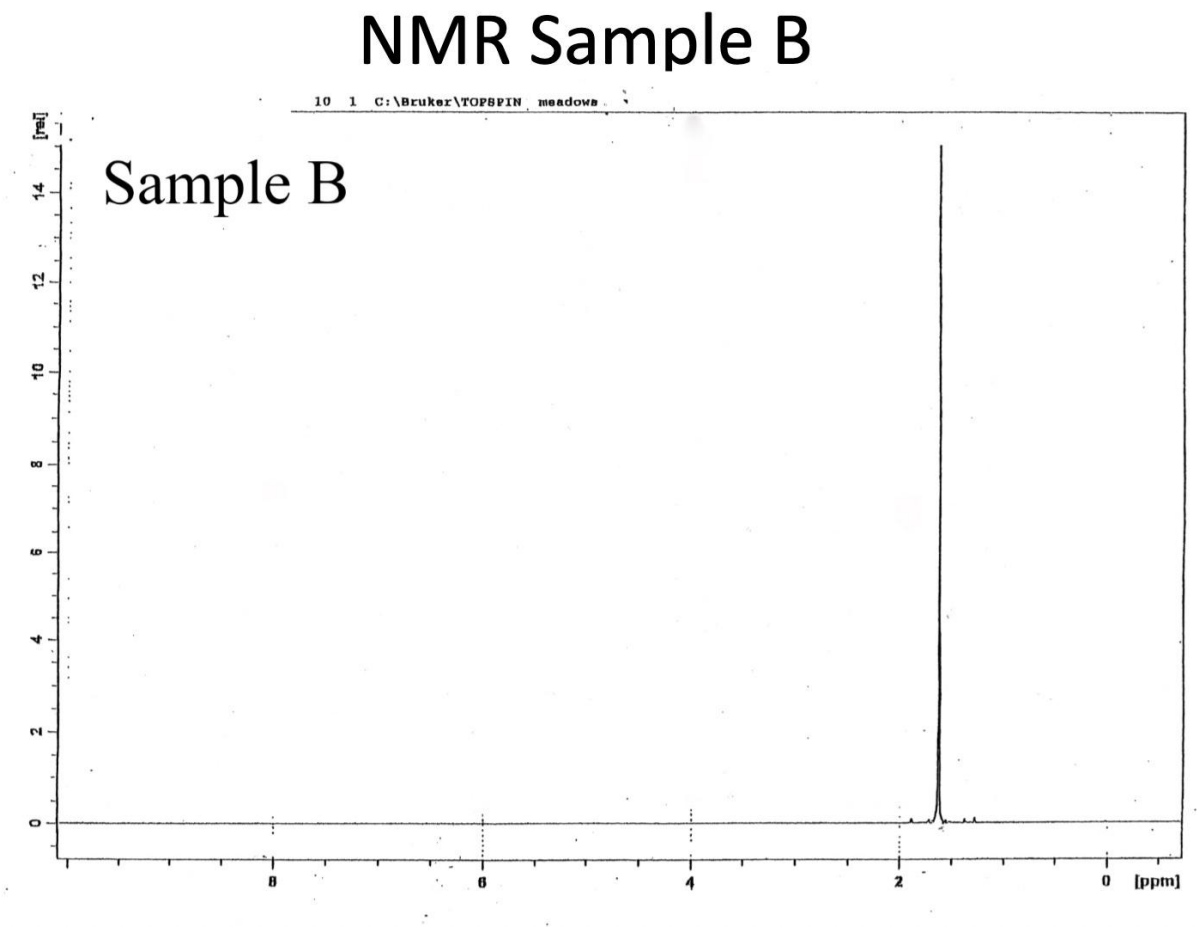
Body

NMR stands for Nuclear Magnetic Resonance. According to the lab manual, the first NMR spectrometers were continuous wave instruments that utilized radio frequency (Ruychev 85). NMR is used by analyzing the interactions between the RF energy and the spin states of the nucleus. NMR differs from MS in that it is used to identify functional groups. The four types of information that can be used to determine the structure of a compound from an NMR spectrum include the number of different kinds of protons, the integration, the chemical shift, and the spin-spin coupling (Ruychev 89).

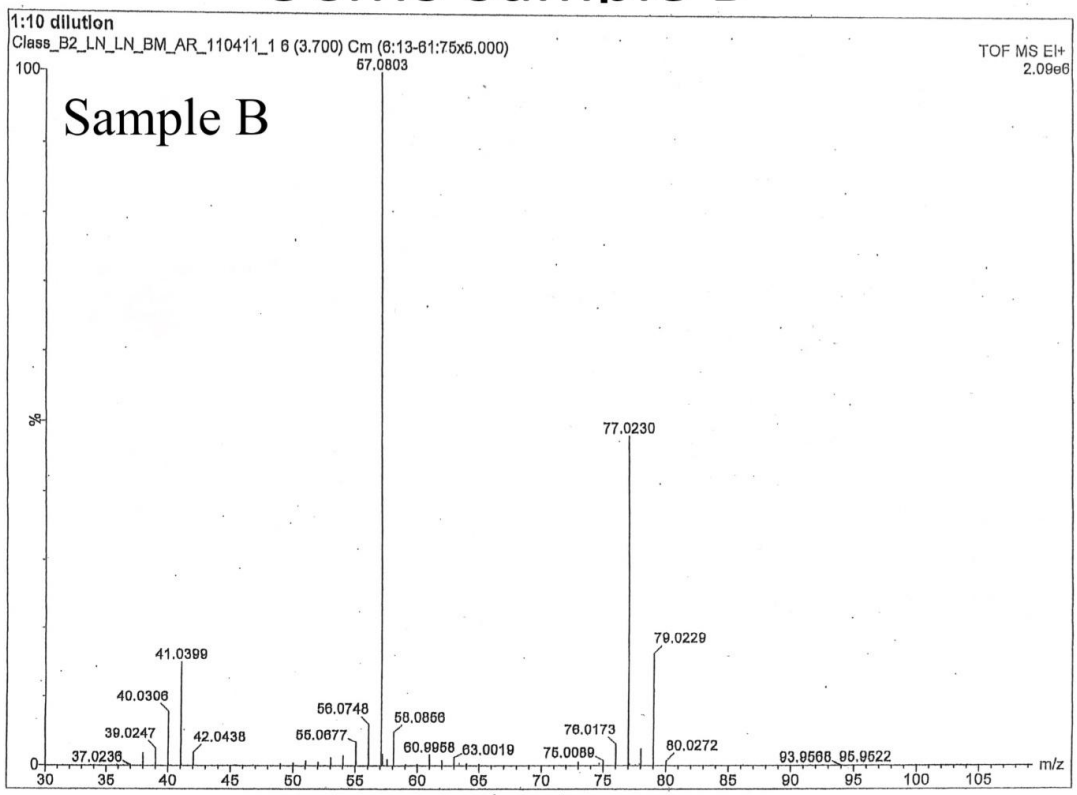
MS stands for Mass Spectroscopy. During MS, a compound is "converted into a gas-phase ion" and a mass analyzer sorts the ions by their mass-to-charge (*m/z*) ratios (Ruychev 157). One benefit to being able to read a MS graph is that it can show the molecular weight of the compound, which is crucial in identifying an unknown. This can be determined if the "molecular ion does not completely fragment before being detected" (Ruychev 161). The *m/z* can also show the chemical formula of a hydrocarbon.

IR stands for infrared spectroscopy. Like NMR, IR can show specific functional groups. Scientists mainly focus on the mid-infrared, which extends from 4000 to 600 cm^{-1} because this is where typical organic compounds appear (Ruychev 133). The location of the absorption bands can provide information on the types of chemical bonds in a compound. The four important factors that can be utilized from an IR spectrum include "the type of vibration, the strength of the bond connecting the atoms, the masses of the atoms attached by the covalent bonds, and the electronegativity difference between the atoms" (Ruychev 139). Fourier Transform Infrared Spectroscopy was used to identify the unknown.

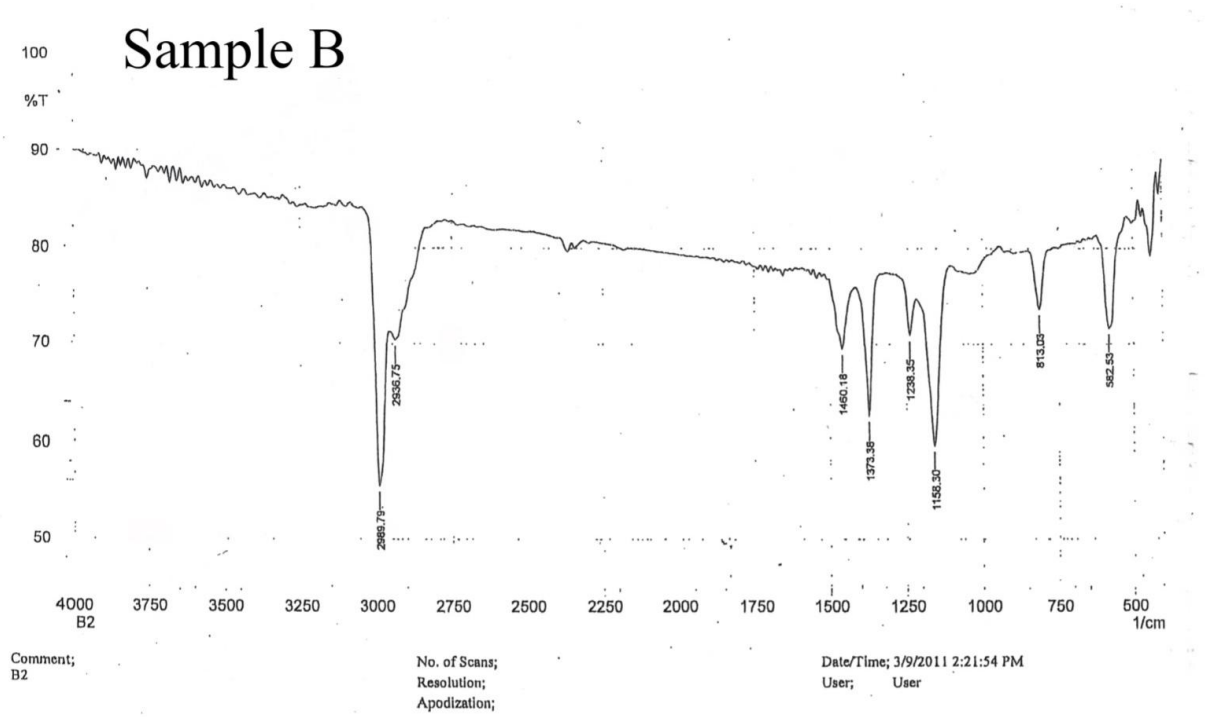
Data



GCMS Sample B



FTIR Sample B



Analysis

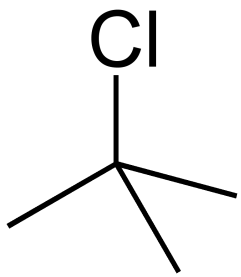
There is a single peak in the NMR graph at 1.6 ppm, indicating a chlorine atom. Since there is only one peak in this graph, it is known that there is only one type of proton in the compound. The height of the signal shows that there are multiple protons. Since the signal is symmetric, the protons are equivalent.

There is a base peak at 57 *m/z* in the MS graph, indicating the formula for the unknown compound is C_4H_9. The molecular ion peak at 93 indicates the molecule tert-butyl chloride. The presence of chlorine discovered in the NMR graph is confirmed by the peaks at 77 and 79 in the MS graph.

The peaks at 2936 and 2989 cm^{-1} indicate single bonds of C-H. Since there are no peaks >3000 cm^{-1}, there are no alcohol groups. The lack of peaks between 1600 and 2700 cm^{-1} show that there are no double or triple bonds. Since there are multiple peaks between 1500 and 1000 cm^{-1}, it is known that there is a methyl group present. The presence of a chlorine atom is further confirmed by the peaks at 582 and 813 cm^{-1}.

Conclusion

The unknown compound B is tert-butyl chloride, also known as 2-Chloro-2-methylpropane. All three graphs indicated the presence of a chlorine atom. All three graphs also indicated a chlorine-carbon bond and the number of carbon and hydrogen atoms present. Therefore, each graph indicated the molecule below.



References

Ruychev, R. Department of Chemistry l Auburn University. In *CHEM 2023 Organic Chemistry II Lab Manual,* 2023 ed.; Macmillan Learning Curriculum Solutions, 2023.